		force exerted $=$ rate of change of momentum of photons $= \frac{\text{no. of photons}}{\text{time}} \quad \frac{2h}{\lambda} = \frac{2nh}{\lambda}$
29	**C**	Statement C is true as it is the result of conservation of linear momentum. The following statements are incorrect: Statement A – The stability of the nucleus depends on the binding energy per nucleon, not the binding energy. Statement B – The reaction can still occur provided energy is given to the reactants. Statement D – The half-life of a radioactive substance is a constant for a given radioactive substance.
30	**D**	Energy released = $BE_{products} - BE_{reactants}$ $= (8.2 \times 146) + (8.6 \times 87) \ldots (7.6 \times 235)$